

## **BACKLOG BUDDY ? STUDY MATERIAL**

Earthquake Resistant Design of Structures ? Unit-wise Lecture Notes

Subject Code: CE602 Semester: 7

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Condensed notes covering all five units: Earthquake Basics and Seismology, Response Spectrum Analysis, Ductile Detailing of RC, Masonry Structures, Base Isolation and Control.

### **Unit 1: Earthquake Basics and Seismology**

- Definition, key concepts and significance of Earthquake Basics and Seismology
- Important diagrams / derivations / algorithms for Earthquake Basics and Seismology
- Solved examples and previous-year questions on Earthquake Basics and Seismology

### **Unit 2: Response Spectrum Analysis**

- Definition, key concepts and significance of Response Spectrum Analysis
- Important diagrams / derivations / algorithms for Response Spectrum Analysis
- Solved examples and previous-year questions on Response Spectrum Analysis

### **Unit 3: Ductile Detailing of RC**

- Definition, key concepts and significance of Ductile Detailing of RC
- Important diagrams / derivations / algorithms for Ductile Detailing of RC
- Solved examples and previous-year questions on Ductile Detailing of RC

### **Unit 4: Masonry Structures**

- Definition, key concepts and significance of Masonry Structures
- Important diagrams / derivations / algorithms for Masonry Structures
- Solved examples and previous-year questions on Masonry Structures

### **Unit 5: Base Isolation and Control**

- Definition, key concepts and significance of Base Isolation and Control
- Important diagrams / derivations / algorithms for Base Isolation and Control
- Solved examples and previous-year questions on Base Isolation and Control

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- \* Revise definitions, diagrams and standard derivations ? they earn easy marks.
- \* Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.